

SIMATS ENGINEERING

ASSESSMENT TOOL 2

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COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4308

1: HTTP Response Analysis

A web server returns the following HTTP response header:

HTTP/1.1 200 OK

Date: Mon, 01 Apr 2026 10:00:00 GMT

Content-Type: text/html

Content-Length: -150

Connection: keep-alive

Questions:

1. Identify any incorrect or invalid fields in the header.

Content-Length: -150 is invalid because Content-Length cannot be negative.
The **Date** format is correct, assuming the date is valid.
Other fields (HTTP/1.1, 200 OK, Content-Type, Connection) are valid.

2. Analyze the issue with Content-Length and explain its impact.

Content-Length specifies the size of the response body in **bytes**.
A negative value is invalid.

Impact:

- a. The browser may reject the response.
- b. The page may fail to load completely.

c. Communication between client and server may become unreliable.

3. Determine whether the status code matches the response correctness.

- **Status Code: 200 OK** means the request was processed successfully.
- Since the response contains an invalid header, the response is **not completely correct**.
- Therefore, the status code does not fully match the response quality.

4. Suggest corrections for the identified errors.

HTTP/1.1 200 OK

Date: Mon, 01 Apr 2026 10:00:00 GMT

Content-Type: text/html

Content-Length: 150

Connection: keep-alive

2: DOM Structure Analysis

Consider the following DOM tree representation:

<html>

<head>

<title>Test Page</title>

</head>

<body>

<div>

<p>Hello World

</div>

</body>

</html>

Questions:

1. Identify missing or improperly closed elements.

Missing `</p>` closing tag.

Missing `</div>` before closing the paragraph (browser automatically fixes it).

2. Analyze how the browser will interpret this structure.

Browsers automatically insert the missing `</p>` before `</div>`.

The page will still display "Hello World".

3. Determine the impact on rendering and layout.

Modern browsers render the page correctly.

Invalid HTML may cause layout issues in complex pages.

JavaScript DOM manipulation may behave unexpectedly.

4. Suggest corrections to make the DOM valid.

Hello World

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL	Example Data	Visibility	Security Level
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GET	/login?	user=admin&pass=1234	Visible in URL	Low
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POST	/login	Hidden in body		Moderate
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Questions:

1. Interpret the differences in data transmission between GET and POST.

GET

- Sends data through the URL.
- Data is visible.
- Used mainly for retrieving information.

POST

- Sends data in the request body.
- Data is not shown in the URL.
- Used for submitting sensitive data.

2. Analyze which method is more suitable for login and why.

POST is better because:

- Username and password are not exposed in the URL.
- More secure than GET.
- Supports larger amounts of data.

3. Identify security risks in using GET for authentication.

Password appears in the URL.
Stored in browser history.
Saved in server logs.
Can be shared accidentally.
Can be bookmarked.

4. Suggest best practices for secure login implementation.

Use **POST** requests.
Always use **HTTPS**.
Hash passwords before storing.
Use secure session management.
Validate user input.
Enable authentication tokens or secure cookies.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph

</body>
</html>
```

Questions:

1. Identify missing closing tags.

Missing:

- `</h1>`
- `</p>`

2. Analyze issues with the `<img>` tag.

- `<img>` is a **void element**, so it does **not** require a closing tag.
- It is better to include an `alt` attribute for accessibility.

Example: ``

3. Determine how browsers will handle these errors.

Browsers automatically close the missing tags.
The page usually displays correctly.
HTML remains invalid and may cause unexpected behavior.

4. Suggest a corrected version of the code.

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome</h1>
<p>This is a paragraph</p>

</body>
</html>
```

5: Browser-Server Communication Flow

The following sequence describes a web request:

1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request
5. Response is returned

6. Browser renders page

Questions:

1. Interpret each step in the communication flow.

Step	Description
1	User types a website address into the browser.
2	DNS converts the domain name into an IP address.

Step	Description
3	Browser sends an HTTP request to the server.
4	Server processes the request and prepares the response.
5	Server sends HTML, CSS, JavaScript, and other resources back to the browser.
6	Browser renders the page for the user to view.

2. Identify where delays can occur.

- Slow DNS lookup.
- Slow internet connection.
- Network latency.
- Busy or overloaded server.
- Large files (images, videos, scripts).
- Slow browser rendering.

3. Analyze the role of DNS in the process.

- DNS translates the website name (e.g., www.example.com) into an IP address.
- Without DNS, the browser cannot locate the web server.
- Faster DNS resolution improves website loading speed.

4. Suggest optimizations to improve performance.

- Use **DNS caching**.
- Enable browser caching.
- Compress files using **Gzip/Brotli**.
- Use a **Content Delivery Network (CDN)**.

- Minify CSS and JavaScript.
- Optimize image sizes.
- Keep server response time low.
- Use persistent connections (keep-alive).